

Chocolate Sales

Salesperson Performance Dashboard

2022 | Power BI Desktop

\$6.18M Revenue · 177,007 Boxes Shipped · 1,094 Records · 6 Countries · 25 Salespersons

TOOL	SOURCE	RECORDS	DATE RANGE
Power BI Desktop	Kaggle (Public)	1,094 · 6 Cols	Jan–Aug 2022

01 EXECUTIVE SUMMARY

- Processed and cleaned 1,094 chocolate sales records spanning 8 months (Jan–Aug 2022) across 25 salespersons, 6 countries, and 22 product categories — eliminating \$ symbols and commas from the Amount column, correcting 2 data type errors, and renaming columns for DAX compatibility before loading into Power BI.
- Engineered 5 DAX measures (Total Revenue, Total Boxes, Avg Revenue per Sale, Total Transactions, Salesperson Rank) and built a 12-visual interactive dashboard featuring KPI cards, leaderboard bar charts, product comparison, monthly trend line, country donut chart, salesperson table, and 2 dynamic slicers.
- Analyzed \$6,183,625 total revenue across 177,007 boxes shipped and 1,094 transactions — surfacing Ches Bonnell as #1 salesperson at \$320,901 (5.19% share) and flagging Wilone O'Kielt at rank 25 with \$138,523 — a \$182,378 performance gap representing a 2.32× revenue difference between top and bottom performer.
- Identified Australia as top revenue country at \$1,137,367 (18.39%), revealed remarkably balanced country distribution (only 3.02pp spread across all 6 countries), flagged Smooth Sliky Salty as top product at \$349,692, and uncovered January as peak revenue month at \$896,105 — 32.9% above the April trough of \$674,051.

02 PROBLEM STATEMENT

Chocolate sales data for 2022 existed as a raw CSV file with 1,094 records and no structured reporting framework. Sales managers had no visibility into which salespersons were consistently outperforming peers, which countries were driving revenue, or which product categories dominated sales — nor was there any mechanism to track monthly revenue trends or identify seasonal demand patterns. Additionally, the raw Amount column contained currency symbols and comma formatting (\$5,320) making it impossible to aggregate revenue without data transformation. Decisions were being made without data-backed insights across a \$6.18M revenue portfolio spanning 6 countries.

03 OBJECTIVES

- Obj 1: Transform raw CSV data into a clean, analysis-ready dataset — removing currency symbols, correcting data types, and renaming columns via Power BI Power Query ETL.
- Obj 2: Engineer 5 DAX measures enabling dynamic aggregation of revenue, boxes, transactions, averages, and salesperson ranking.

- Obj 3: Build a 12-visual interactive dashboard — salesperson leaderboard, country comparison, product breakdown, monthly trend, donut chart, and performance table.
- Obj 4: Enable cross-filtering via 2 dynamic slicers (Country, Product) for interactive multi-dimensional exploration.
- Obj 5: Surface actionable insights — top/bottom salespersons, country distribution, product rankings, and seasonal revenue patterns.
- Obj 6: Deliver a portfolio-ready report with a custom chocolate-branded JSON theme matching the dataset's visual identity.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Chocolate Sales 2022
Source	Kaggle — atharvasoundankar/chocolate-sales
File Format	CSV (.csv)
Total Records	1,094 rows
Total Columns	6 (after renaming 'Sales Person' to 'Salesperson')
Date Coverage	January 3, 2022 – August 31, 2022 (8 months)
Geographic Scope	6 countries: Australia, UK, India, USA, Canada, New Zealand
Null Values	Zero — 100% complete dataset
Data Issue	Amount column stored as text with \$ and commas (\$5,320) — required Power Query fix

Column Definitions

Column	Type	Description
Salesperson	Text	25 unique sales representatives across all 6 countries
Country	Text	6 countries: Australia, UK, India, USA, Canada, New Zealand
Product	Text	22 chocolate product categories (e.g. Smooth Sliky Salty, 50% Dark Bites)
Date	Date	Transaction date — daily granularity, Jan–Aug 2022
Amount	Decimal	Revenue per transaction in USD (originally stored as text '\$5,320')
Boxes Shipped	Whole Number	Units shipped per transaction (avg 161.8 boxes per sale)

05 TOOLS & TECHNOLOGIES USED

Tool / Technology	Platform	Purpose in This Project
Power BI Desktop	Microsoft (Latest 2024)	Primary BI tool — ETL, DAX, data modelling, and dashboard visualisation
Power Query Editor	Built into Power BI	ETL pipeline — \$ removal, comma removal, type correction, column rename
DAX (Data Analysis Expressions)	Power BI Native	5 measures: Revenue, Boxes, Avg Sale, Transactions, Salesperson Rank
CSV Source File	Comma-Separated Values (.csv)	Raw data format — imported via Get Data → Text/CSV connector
Kaggle	Public Dataset Platform	Free public dataset — atharvasoundankar/chocolate-sales
JSON Theme File	Power BI Custom Theme	Chocolate Indulgence — dark brown/caramel/gold branded theme

06 ETL PROCESS — DATA PREPARATION IN POWER QUERY

The key ETL challenge was the Amount column — stored as formatted text ('\$5,320') rather than numeric values. Power BI cannot perform SUM or AVERAGE on text fields, requiring two Replace Values steps to strip the \$ symbol and thousand-separator comma before converting to Decimal Number. All 7 steps were executed in Power Query Editor and are logged in the Applied Steps pane for full auditability.

Step	Power Query Action	Reason / Impact
01	Get Data → Text/CSV → selected Chocolate_Sales.csv	Loaded raw CSV with preview showing all 6 columns
02	Clicked 'Transform Data' to open Power Query Editor	Enabled pre-load cleaning before data enters model
03	Amount column → Replace Values: '\$' → " (empty)	Removed dollar sign from all 1,094 Amount values
04	Amount column → Replace Values: ',' → " (empty)	Removed thousand-separator commas (e.g. 5,320 → 5320)
05	Amount column type → Decimal Number	Converted text to numeric — enabled SUM/AVERAGE DAX measures
06	Date column type → Date	Converted text dates ('04-Jan-22') to proper Date type for time charts
07	Renamed 'Sales Person' → 'Salesperson'	Removed space — cleaner field names for DAX and visual labels
08	Home → Close & Apply	Loaded 1,094 clean rows — 0 errors, 0 nulls, all types correct

07 DAX MEASURES CREATED

Measure Name	DAX Formula	Result / Use
Total Revenue	SUM('Chocolate Sales'[Amount])	\$6,183,625 — primary revenue KPI
Total Boxes	SUM('Chocolate Sales'[Boxes Shipped])	177,007 — volume KPI card
Avg Revenue per Sale	AVERAGE('Chocolate Sales'[Amount])	\$5,652 — avg transaction value
Total Transactions	COUNTROWS('Chocolate Sales')	1,094 — total transaction count
Salesperson Rank	RANKX(ALL('Chocolate Sales'[Salesperson]),[Total Revenue],,DESC)	Dynamic rank #1–25 per filter context

08 DASHBOARD VISUALS BUILT (12 Visuals)

#	Visual Type	Fields Used	Insight Delivered
1	KPI Card	Total Revenue	\$6,183,625 total chocolate sales revenue
2	KPI Card	Total Boxes	177,007 total boxes shipped
3	KPI Card	Total Transactions	1,094 total sales transactions
4	KPI Card	Avg Revenue per Sale	\$5,652 average transaction value
5	Clustered Bar Chart	Y: Salesperson / X: Total Revenue	Ches Bonnell #1 at \$320,901 leaderboard
6	Clustered Bar Chart	Y: Country / X: Total Revenue	Australia leads at \$1,137,367 (18.39%)
7	Clustered Bar Chart	Y: Product / X: Total Revenue	Smooth Sliky Salty #1 at \$349,692
8	Line Chart	X: Date / Y: Total Revenue	Monthly trend Jan peak (\$896K) to Apr trough (\$674K)
9	Donut Chart	Legend: Country / Values: Revenue	Balanced country share — 15.37%–18.39% spread
10	Table	Salesperson, Revenue, Boxes, Transactions, Rank	Full 25-person leaderboard with rank
11	Slicer	Country (6 values)	Filters all visuals by country
12	Slicer	Product (22 values)	Filters all visuals by product

09 KEY INSIGHTS & RESULTS

A. Revenue & Volume Overview

- Analyzed \$6,183,625 total revenue across 1,094 transactions and 177,007 boxes shipped — averaging \$5,652.31 per sale and 161.8 boxes per transaction across 25 salespersons, 6 countries, and 22 products over 8 months (Jan–Aug 2022).

- Identified January as peak revenue month at \$896,105 and June as second-highest at \$865,144 — with April as the seasonal trough at \$674,051, representing a 24.8% revenue dip that recovered fully by May (\$752,892).

B. Salesperson Leaderboard

- Ches Bonnell ranked #1 at \$320,901 (5.19% share, 7,522 boxes, 48 transactions); Oby Sorrel #2 at \$316,645; Madelene Upcott #3 at \$316,099 — top 3 tightly clustered within \$4,802 of each other, indicating strong competitive performance.
- Wilone O'Kielt ranked last at #25 with \$138,523 (2.24% share, 4,033 boxes, 34 transactions) — \$182,378 below Ches Bonnell and generating only 43.2% of the top performer's revenue, flagging significant performance gap requiring sales coaching intervention.
- Kelci Walkden led all salespersons in transactions at 54 despite ranking only #5 in revenue at \$311,710 — indicating high volume at lower average deal size (\$5,772/transaction vs Ches Bonnell's \$6,686) suggesting opportunity for upselling.

C. Country Performance

- Australia ranked #1 at \$1,137,367 (18.39%); UK #2 at \$1,051,792 (17.01%); India #3 at \$1,045,800 (16.91%); USA #4 at \$1,035,349 (16.74%); Canada #5 at \$962,899 (15.57%); New Zealand #6 at \$950,418 (15.37%).
- Remarkably balanced country distribution — only 3.02 percentage point spread between top (Australia 18.39%) and bottom (New Zealand 15.37%) country — confirming globally consistent sales execution with no single market dependency or concentration risk.

D. Product Analysis

- Smooth Slikly Salty ranked #1 product at \$349,692 (5.66% share, 8,810 boxes); 50% Dark Bites #2 at \$341,712 (5.53%); White Choc #3 at \$329,147 (5.32%) — top 3 products contributing \$1,020,551 combined (16.51% of total revenue).
- 70% Dark Bites ranked last at \$211,610 (3.42%); Caramel Stuffed Bars #21 at \$231,588 (3.75%) — representing \$138,082 revenue gap between #1 and #22 product. All 22 products remained active with no zero-revenue SKUs across the 8-month period.

E. Salesperson Performance Table (All 25)

Rank	Salesperson	Revenue	Boxes	Transactions	Rev Share%
1	Ches Bonnell	\$320,901	7,522	48	5.19%
2	Oby Sorrel	\$316,645	8,608	49	5.12%
3	Madelene Upcott	\$316,099	7,279	45	5.11%
4	Brien Boise	\$312,816	8,102	53	5.06%
5	Kelci Walkden	\$311,710	8,702	54	5.04%
6	Van Tuxwell	\$303,149	6,799	51	4.90%
7	Dennison Crosswaite	\$291,669	8,767	49	4.72%

Rank	Salesperson	Revenue	Boxes	Transactions	Rev Share%
8	Beverie Moffet	\$278,922	9,214	50	4.51%
9	Kaine Padly	\$266,490	7,253	45	4.31%
10	Marney O'Brien	\$259,742	8,043	45	4.20%
11	Barr Faughny	\$258,713	6,366	43	4.18%
12	Roddy Speechley	\$251,062	6,899	43	4.06%
13	Gunar Cockshoot	\$238,483	6,677	43	3.86%
14	Gigi Bohling	\$232,666	6,303	47	3.76%
15	Karlen McCaffrey	\$223,895	9,658	47	3.62%
16	Jehu Rudeforth	\$220,976	7,246	43	3.57%
17	Jan Morforth	\$219,667	7,661	39	3.55%
18	Curtice Advani	\$216,461	7,074	46	3.50%
19	Rafaelita Blakslund	\$210,245	4,297	34	3.40%
20	Husein Augar	\$205,212	5,849	38	3.32%
21	Andria Kimpton	\$201,747	6,448	39	3.26%
22	Mallorie Waber	\$200,592	5,980	41	3.24%
23	Camilla Castle	\$196,616	5,374	32	3.18%
24	Dotty Strutley	\$190,624	6,853	36	3.08%
25	Wilone O'Kielt	\$138,523	4,033	34	2.24%

10 KPI SUMMARY TABLE

KPI	Value	KPI	Value
Total Revenue	\$6,183,625	Total Transactions	1,094
Total Boxes Shipped	177,007	Avg Revenue per Sale	\$5,652.31
Unique Salespersons	25	Avg Boxes per Sale	161.8
Unique Countries	6	Unique Products	22
#1 Salesperson	Ches Bonnell — \$320,901	#25 Salesperson	Wilone O'Kielt — \$138,523
Performance Gap	\$182,378 (2.32×)	Top Country	Australia — \$1,137,367 (18.39%)
Country Spread	3.02pp (18.39%–15.37%)	Top Product	Smooth Sliky Salty — \$349,692
Peak Month	January — \$896,105	Trough Month	April — \$674,051
Monthly Range	\$896,105–\$674,051	Revenue Seasonality	24.8% peak-to-trough dip

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Amount Column Stored as Currency Text

- Problem: The Amount column contained values formatted as '\$5,320' (text string with dollar sign and comma) making it impossible to use SUM() or AVERAGE() DAX functions — attempting to aggregate produced blank or error results.
- Solution: Applied 2 sequential Replace Values steps in Power Query — first replacing '\$' with empty string, then ',' with empty string — converting '\$5,320' to '5320'. Then changed column type to Decimal Number, enabling all revenue calculations instantly.

Challenge 2 — Date Stored as Text String

- Problem: Dates were stored as text in '04-Jan-22' format. Without correction, Power BI could not create date hierarchies (Year/Quarter/Month/Day), preventing the line chart from grouping data by month or showing proper time-axis labels.
- Solution: Changed Date column type to 'Date' in Power Query. Power BI automatically parsed the 'DD-Mon-YY' format into proper date values, enabling the line chart to display monthly grouping and drill-down from Year → Quarter → Month → Day.

Challenge 3 — Column Name with Space Causing DAX Issues

- Problem: The original column was named 'Sales Person' (with a space). While Power BI can handle spaces in column names, it requires single-quote wrapping in every DAX formula ('Chocolate Sales'[Sales Person]) — error-prone and harder to read.
- Solution: Renamed 'Sales Person' → 'Salesperson' (no space) in Power Query via double-click on column header. Eliminated the space, making all DAX formulas cleaner and reducing chance of syntax errors throughout the project.

Challenge 4 — RANKX Measure Showing Same Rank for All Rows

- Problem: Initial RANKX formula didn't include ALL() function — causing the rank to always show '1' for every salesperson because it was ranking within the already-filtered context, not across all salespersons.
- Solution: Added ALL('Chocolate Sales'[Salesperson]) as the first argument: RANKX(ALL('Chocolate Sales'[Salesperson]), [Total Revenue], , DESC). The ALL() removes the filter context, forcing RANKX to rank across all 25 salespersons regardless of slicer selections.

Challenge 5 — Line Chart Showing Daily Spikes Instead of Monthly Trend

- Problem: The line chart showed hundreds of noisy daily data points instead of a clean monthly trend — making it impossible to identify seasonal patterns or month-over-month changes in revenue.
- Solution: Clicked the X-axis drill-up button on the line chart to collapse from Day level to Month level. Alternatively, right-clicked the Date hierarchy on the X-axis and selected 'Month' level — creating a clean 8-point monthly trend line from January to August 2022.

12 SKILLS LEARNED & TAKEAWAYS

- Currency Text Cleaning: Mastered multi-step Replace Values in Power Query to strip currency symbols and thousand separators — a real-world ETL challenge found in virtually every financial dataset.
- Advanced DAX — RANKX(): Implemented dynamic salesperson ranking using RANKX with ALL() to override filter context — understanding how ALL() enables cross-context calculations that respond correctly to slicer interactions.
- COUNTROWS() vs COUNT(): Used COUNTROWS() for Total Transactions measure — learning that COUNTROWS() counts table rows while COUNT() counts non-blank values in a column, with COUNTROWS() being more reliable for transaction counting.
- Date Hierarchy Drill-Down: Learned to control line chart date granularity using the drill-up/drill-down buttons, switching between Year/Quarter/Month/Day views to find the right level of trend visibility.
- Leaderboard Design: Built a ranked salesperson leaderboard combining Bar Chart (visual ranking) with Table visual (precise numbers + rank measure) — understanding when each visual type adds unique value to the same underlying data.
- Balanced Distribution Recognition: Identified the 3.02pp country spread as a notable business insight — a near-perfectly balanced global revenue distribution that signals consistent international sales execution across all 6 territories.
- Custom Branded Theming: Created a chocolate-themed JSON file (dark brown canvas, gold accents, caramel data colors) — applying real brand-matching design thinking to data dashboard presentation.

13 CONCLUSION

- Processed 1,094 chocolate sales records through a structured 8-step Power Query ETL pipeline — stripping currency symbols and commas from Amount column, correcting 2 data types, renaming 1 column — achieving 100% clean data load with zero nulls across all 6 fields.
- Engineered 5 DAX measures including a dynamic RANKX()-based Salesperson Rank and built a 12-visual interactive dashboard spanning KPI cards, leaderboard charts, product comparison, monthly trend, country donut, performance table, and 2 cross-filtering slicers.
- Surfaced \$6,183,625 total revenue, Ches Bonnell as #1 salesperson (\$320,901), a \$182,378 performance gap to the bottom performer, Australia as top country (\$1,137,367 — 18.39%), and a remarkably balanced 3.02pp spread across all 6 global markets.
- Identified January as peak revenue month (\$896,105), a 24.8% seasonal dip to April (\$674,051), Smooth Sliky Salty as top product (\$349,692), and Kelci Walkden as highest-volume salesperson (54 transactions) — actionable insights for sales planning and incentive strategy.
- Delivered a custom chocolate-branded Power BI JSON theme (dark brown canvas, gold card borders, caramel data colors) — demonstrating end-to-end command of Power BI from CSV ingestion through ETL, DAX modelling, visual design, and portfolio-quality brand theming. Project 2 of 9.